

**YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCE
TOPIC TEST**

Name: _____

Mark: 53**Part 1: Multiple Choice**

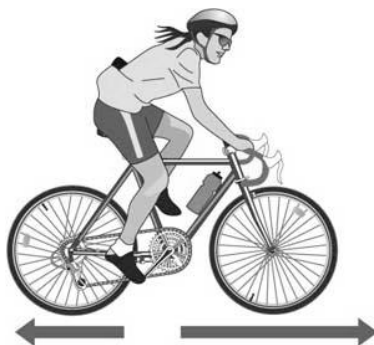
Write your choice for each question in the table below.

1. The crumple zone of a car is designed to:
 - (a) allow the vehicle and passengers to stop more slowly.
 - (b) decrease the time before the car comes to a stop after a collision.
 - (c) allow the chassis to bend.
 - (d) create a reaction force that equals the action force.
2. How long would a car travelling at a speed of 20 m/s take to cover a distance of 360 m?
 - (a) 13 s
 - (b) 26 s
 - (c) 18 s
 - (d) 34 s
3. A car manufacturer wants to produce cars that accelerate faster. Which changes in their cars would result in greater acceleration?
 - (a) Increase both the forward force by the engine and the mass of the car.
 - (b) Decrease both the forward force and the mass of the car.
 - (c) Increase the forward force and decrease the mass of the car.
 - (d) Decrease the forward force and increase the mass of the car.
4. Speed is a measure of the rate:
 - (a) at which distance is covered.
 - (b) at which speed increases.
 - (c) of change in force.
 - (d) of change in acceleration.
5. The rate of change of speed of an object is known as:
 - (a) average speed.
 - (b) average velocity.
 - (c) change in velocity.
 - (d) acceleration.
6. An object starts from rest and accelerates at 5.0 m/s^2 . After 4.0 seconds, its speed will be:
 - (a) 0
 - (b) 10 m/s.
 - (c) 20 m/s.
 - (d) 30 m/s.

MULTIPLE CHOICE ANSWERS	
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7. The tendency of an object to resist changes in its motion is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.
8. Newton's law, known as the law of inertia, is:
- (a) his first law.
 - (b) his second law.
 - (c) his third law.
 - (d) none of the above.
9. Belinda walks a total distance of 1200 metres to school each morning. If this trip takes her 800 seconds, her average speed would be:
- (a) 1.5 m/s.
 - (b) 0.67 m/s.
 - (c) 0 m/s.
 - (d) 960,000 m/s.
10. An object is acted upon by a force of 50 N pushing it forwards and a total frictional force of 40 N acting in the opposite direction. The nett force will be:
- (a) 10 N backwards.
 - (b) 10 N forwards.
 - (c) 90 N backwards.
 - (d) 90 N forwards.
11. The size of the acceleration acting on an object of mass 10 kg that has a nett force of 3.0 N acting on it will be:
- (a) 13 m/s².
 - (b) 3.33 m/s².
 - (c) 0.3 m/s².
 - (d) 30 m/s².
12. A car is driven along a highway at an average speed of 90 km/h for 2 hours without a rest stop. What distance has it covered?
- (a) 45 km
 - (b) 180 km
 - (c) 360 km
 - (d) Not enough information is given to calculate.
13. The force of gravity acting on an object is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.

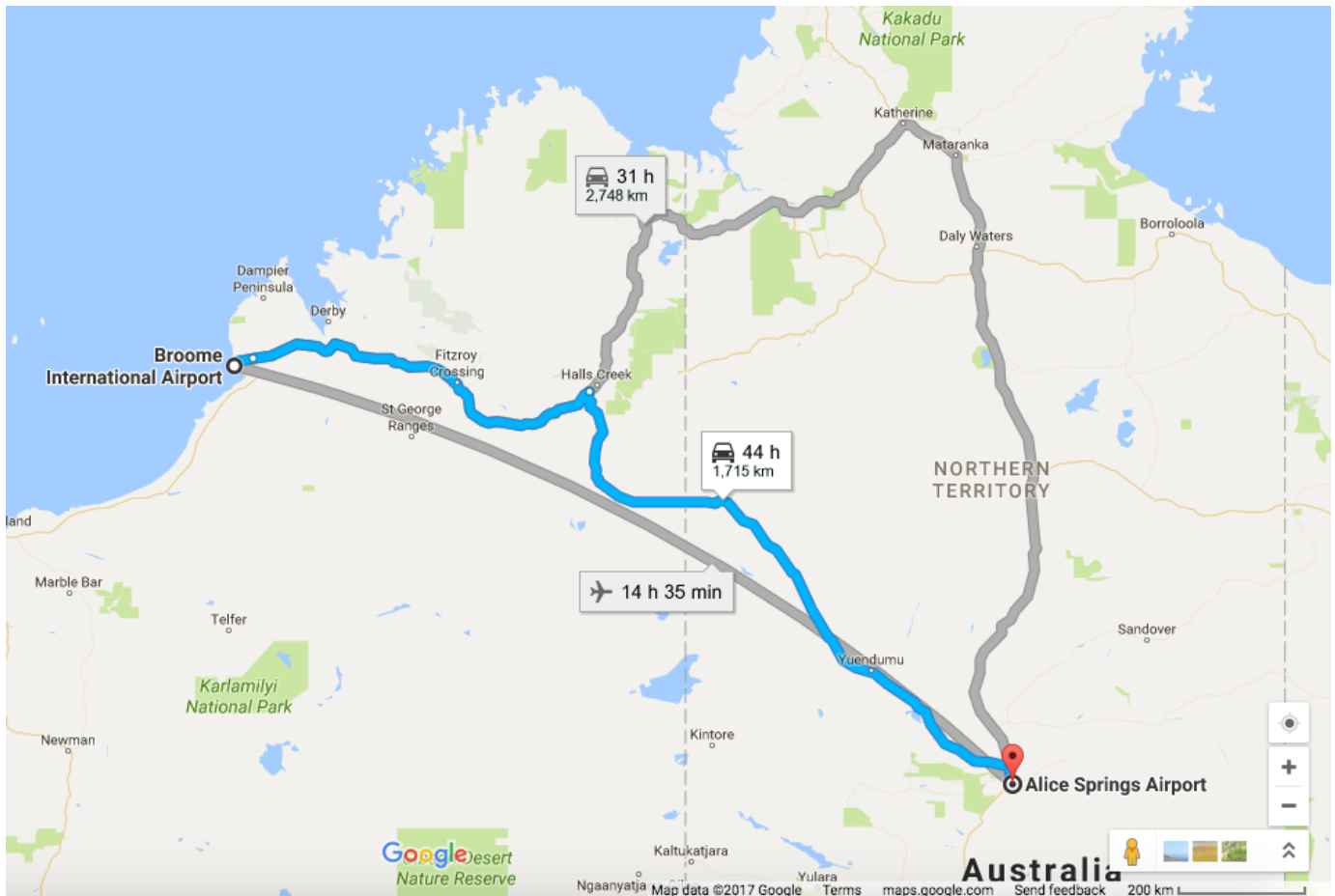
- 14 The weight force acting on a 20 kg mass on the Earth would be:
- (a) 19.8 N
 - (b) 198 N
 - (c) 1.98 N
 - (d) 0 N.
15. In the diagram below, the arrows show the size and direction of two force arrows acting on the bike.



- Which of the following describes the effect of the pair of forces on the bike?
- (a) The bike will accelerate forward.
 - (b) The bike will change its direction of motion.
 - (c) The bike will remain stationary.
 - (d) The bike will decelerate.
16. What is the acceleration of a car if it increases its velocity from 5.0 m/s to 15.0 m/s in 4.0 seconds?
- (a) 2.5 m/s
 - (b) 15 m/s²
 - (c) 2.5 m/s²
 - (d) 5.2 m/s²
17. A class is studying Newton's three laws. The teacher asks a student wearing rollerblades to face the wall and push against it. The student moves backwards away from the wall. Which is the best explanation for her movement?
- (a) Her action of pushing resulted in a reaction force by the wall, in the opposite direction to her push.
 - (b) She used her feet to push herself away from the wall.
 - (c) The push is a force and causes her to accelerate.
 - (d) The inertia of the student is responsible for her acceleration.
18. Which of the following statements is **correct**?
- (a) A car at rest has at least two forces acting on it.
 - (b) A car at rest has no forces acting on it.
 - (c) When you take your foot off the car's accelerator, the car slows down because there are no more forces acting on it.
 - (d) When you brake in a car that is rolling down a hill, the car stops because all of the forces add up to zero.

Part 2: Short Answer

1. Decide whether each of the following statements is **true (T)** or **false (F)**. (4 marks)
 - (a) Average speed is calculated by distance covered multiplied by the time taken. _____
 - (b) $40.0 \text{ km/h} = 144 \text{ m/s}$ _____
 - (c) When a car accelerates quickly, you move forward in your seat. _____
 - (d) A negative acceleration occurs when a car goes backwards. _____
2. The map below shows Broome and Alice Springs. Use the map and its scale to work out the following.
 - (a) The **distance** between Broome Airport and Alice Springs airport. (1 mark)
 - (b) The **displacement** between the two airports. (HINT: Don't forget to include a **bearing**.) (3 marks)



3. Complete the following table on **speed** calculations. Make sure you include the appropriate units for each quantity.

(3 marks)

Distance travelled	Time taken	Average speed
490 m	35 s	
	12 s	23 m/s
121 m		11 m/s

4. Explain why it is dangerous to have large parcels or sharp objects loose in the back of a moving car.

(2 marks)

5. Describe how a front air bag protects the driver of a car that has stopped suddenly as a result of a collision.

(2 marks)

6. Describe the motion of a ball placed on the middle of the back seat of a car in the following situations.

(a) The car brakes suddenly. _____ (1 mark)

(b) The car turns right. _____ (1 mark)

(c) The car accelerates. _____ (1 mark)

7. A dragster accelerates from rest at 14.5 m/s^2 for 6.90 s during a race. What is the velocity of the vehicle after this time? (***Set your working out clearly.***) (3 marks)

8. Complete the following table on ***force*** and ***acceleration***. (3 marks)

Force	Mass	Acceleration
	1200 kg	5 m/s^2
150 N	25 kg	
60 N		2.5 m/s^2

9. The Space Shuttle lands at 270 km/h (75 m/s) and immediately pops a drag chute to assist its brakes to stop the vehicle. Assume that the braking forces are uniform for this motion.



- (a) If it takes 15 s to stop, calculate the **deceleration** experienced by the crew. (3 marks)
- (b) If the mass of the Shuttle is 34,500 kg, what is the average **force** exerted by the brakes and the chute?
(If you couldn't find an answer for part (a), use deceleration = - 6.0 m/s².) (3 marks)

10. An object has a mass of 6.0 kg.

(a) What is its **weight** on Earth?

(2 marks)

(b) Would the object **weigh more or less** on the Moon? Explain your answer.

(2 marks)

(c) What is its **mass** on the Moon?

(1 mark)